02 GROUND CANCEL BUFFER, MIX L

File: master_sheets/02_ground_cancel_buffer.kicad_sch

03 MIX L, MIX R

File: master_sheets/03_mix_l.kicad_sch

04 MIX R, MASTER FADER

File: master_sheets/04_mix_r.kicad_sch

05 MASTER FADER, MONO

File: master_sheets/05_master_fader.kicad_sch

06 MAIN OUTPUTS

File: master_sheets/06_main_outputs.kicad_sch

07 MAIN OUTPUTS, POWER-ON MUTE

File: master_sheets/07_main_outputs.kicad_sch

08 POWER-ON MUTE, AUX MASTER 5

File: master_sheets/08_power_on_mute.kicad_sch

09 AUX MASTER 5

File: master_sheets/09_aux_master_5.kicad_sch

10 AUX MASTER 6

File: master_sheets/10_aux_master_6.kicad_sch

11 AUX MASTER 6, EXT INPUT

File: master_sheets/11_aux_master_6.kicad_sch

12 EXT INPUT, PFL-AFL

File: master_sheets/12_ext_input.kicad_sch

13 PFL-AFL, PFL SWITCHER

File: master_sheets/13_pfl_afl.kicad_sch

14 PFL SWITCHER, PFL DETECT

File: master_sheets/14_pfl_switcher.kicad_sch

15 PFL DETECT, RUDE SOLO

File: master_sheets/15_pfl_detect.kicad_sch

16 CONTROL ROOM

File: master_sheets/16_control_room.kicad_sch

17 CONTROL ROOM, HEADPHONES

File: master_sheets/17_control_room.kicad_sch

18 HEADPHONES, METER L

File: master_sheets/18_headphones.kicad_sch

19 METER L

File: master_sheets/19_meter_l.kicad_sch

20 METER L

File: master_sheets/20_meter_l.kicad_sch

21 METER L

File: master_sheets/21_meter_l.kicad_sch

22 METER L, METER R

File: master_sheets/22_meter_l.kicad_sch

23 METER R

File: master_sheets/23_meter_r.kicad_sch

24 METER R

File: master_sheets/24_meter_r.kicad_sch

25 METER R

File: master_sheets/25_meter_r.kicad_sch

26 JACK LINK CLAMPS

File: master_sheets/26_jack_link_clamps.kicad_sch

27 JACK LINK CLAMPS, POWER ENTRY

File: master_sheets/27_jack_link_clamps.kicad_sch

28 DISTRIBUTION

File: master_sheets/28_distribution.kicad_sch

29 MOTHERLOOM, REMOTE METER HEADER

File: master_sheets/29_motherloom.kicad_sch

30 JACK BOARD LINK

File: master_sheets/30_jack_board_link.kicad_sch

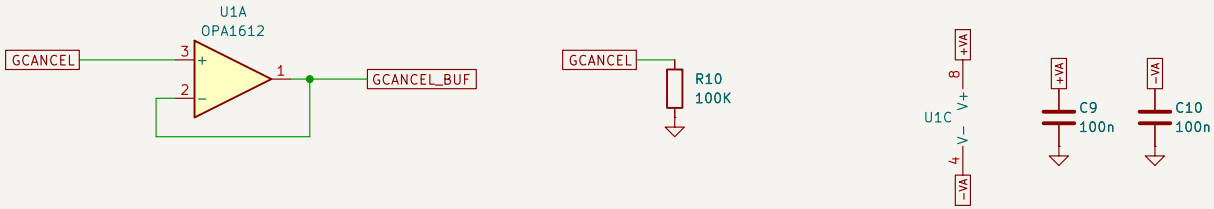
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9

Sheet: /
File: master.kicad_sch

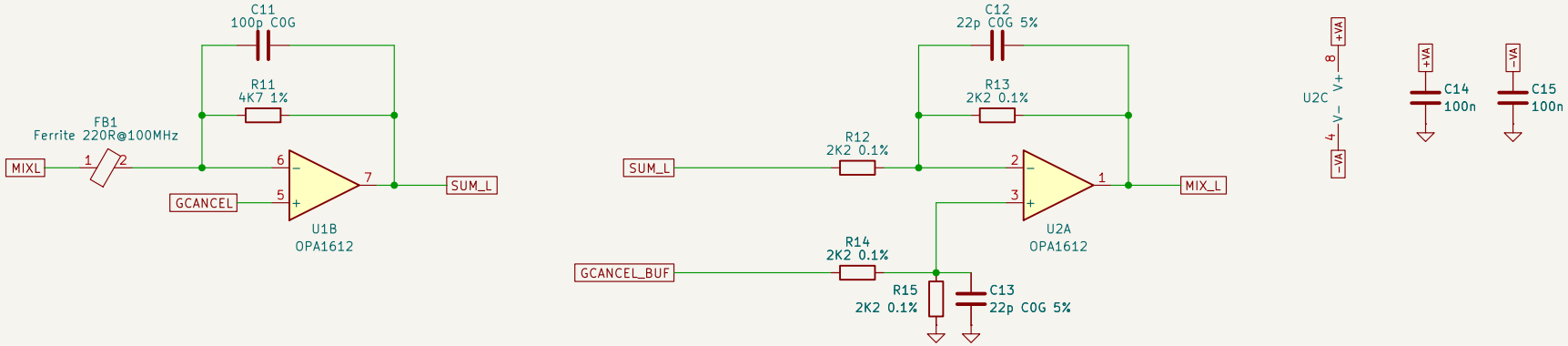
**Title: Master section — goa console**

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/30 |

## GROUND CANCEL BUFFER  (feeds every re-inverter; summing amps use the raw GCANCEL line; 100K keeper)




## MIX L: summing amp (GCANCEL-referenced), re-inverter, master insert



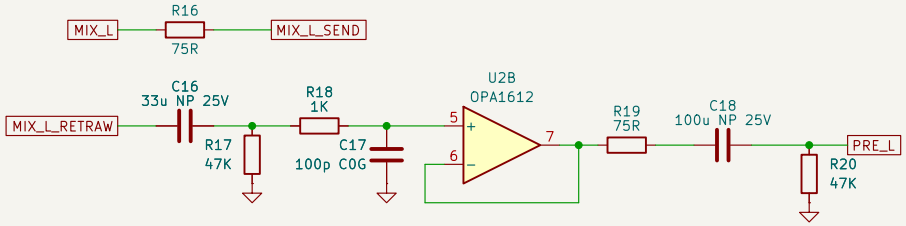

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /02 GROUND CANCEL BUFFER, MIX L/
File: 02_ground_cancel_buffer.kicad_sch

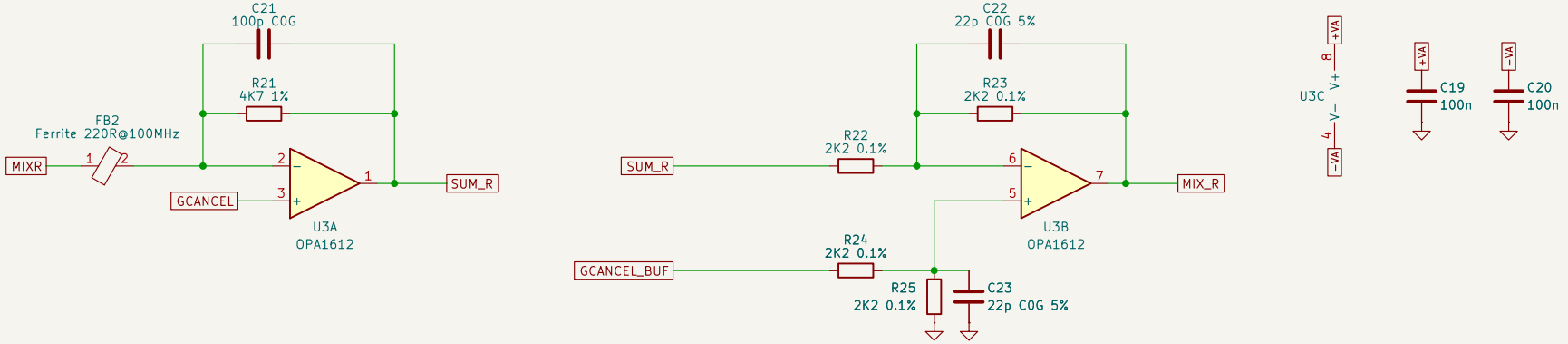
***Title: GROUND CANCEL BUFFER, MIX L***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 2/30 |

## MIX L: summing amp (GCANCEL-referenced), re-inverter, master insert (cont.)




## MIX R: summing amp (GCANCEL-referenced), re-inverter, master insert







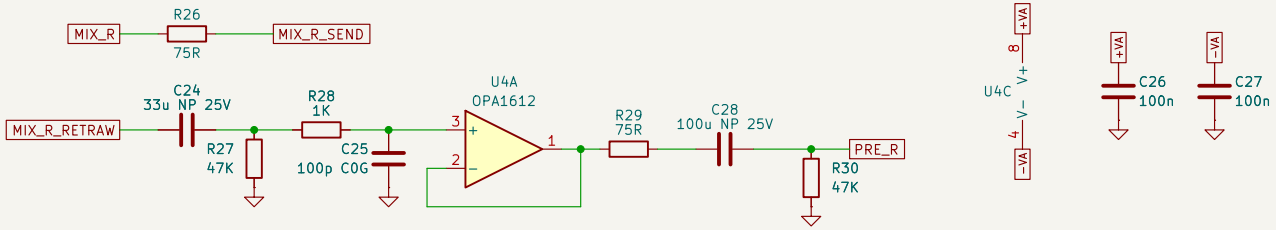
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /03 MIX L, MIX R/
File: 03_mix_l.kicad_sch

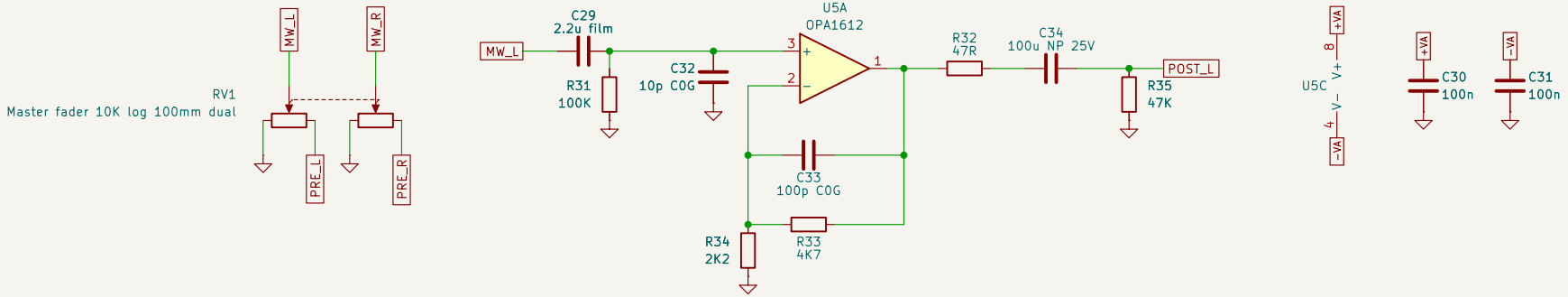
***Title: MIX L, MIX R***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 3/30 |

## MIX R: summing amp (GCANCEL-referenced), re-inverter, master insert (cont.)




## MASTER FADER (dual 100 mm A10K, pin 1 = GNDA) + post-amps +10 dB



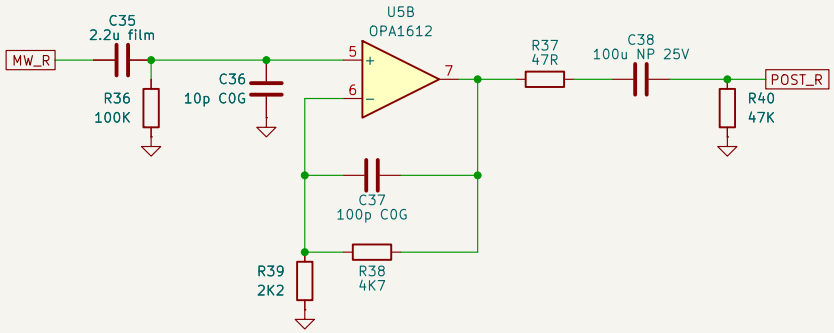

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /04 MIX R, MASTER FADER/
File: 04_mix_r.kicad_sch

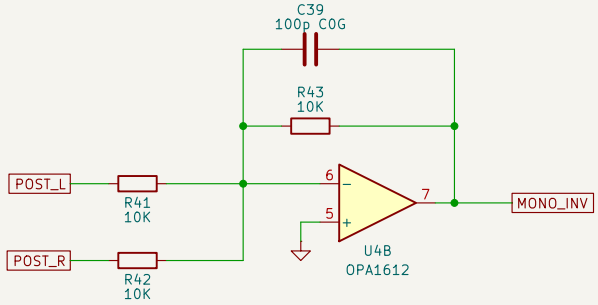
**Title: MIX R, MASTER FADER**

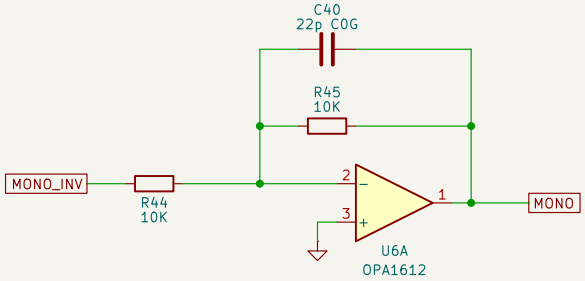
| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 4/30 |

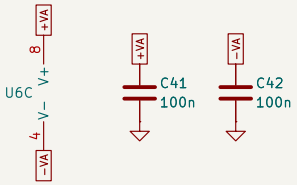
## MASTER FADER (dual 100 mm A10K, pin 1 = GNDA) + post-amps +10 dB (cont.)

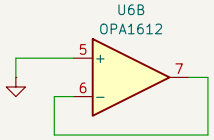



## MONO: inverting sum of POST L+R, re-inverter



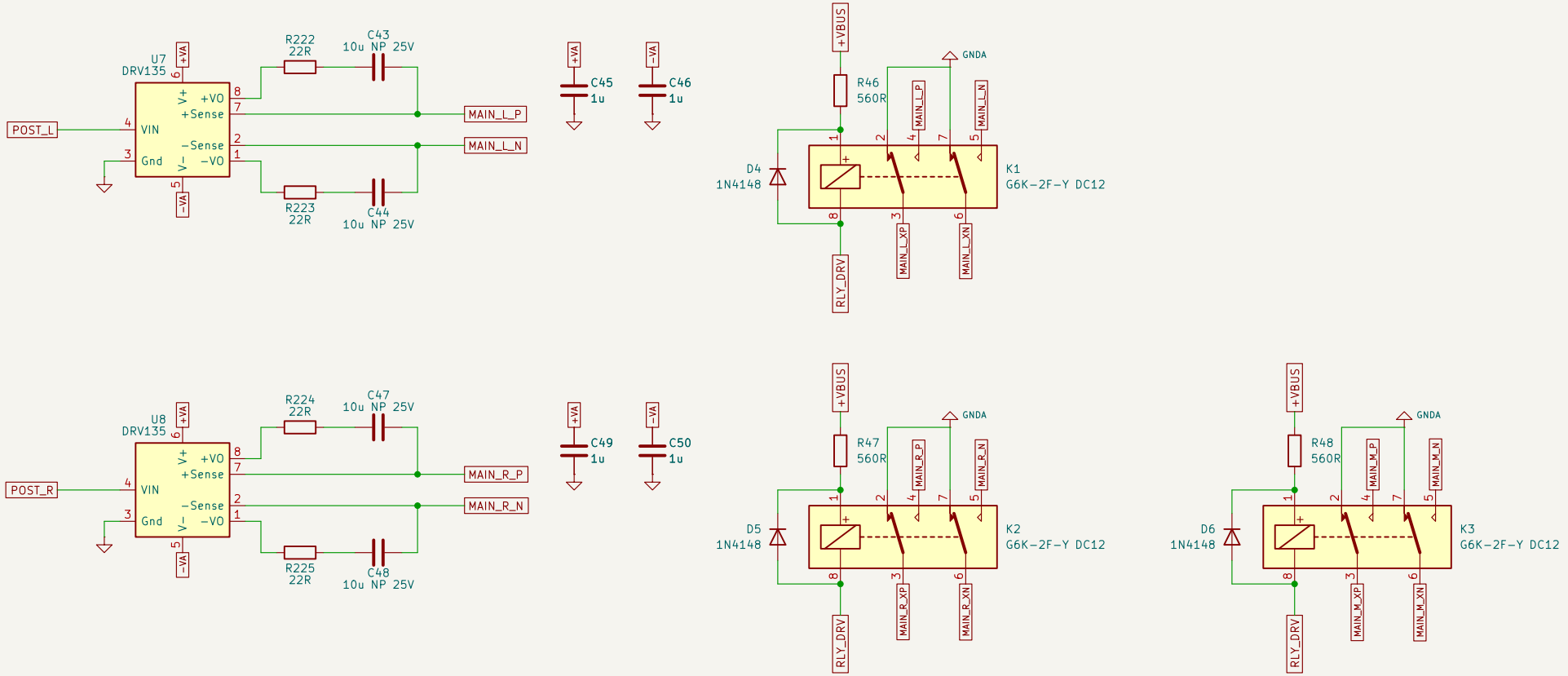

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /05 MASTER FADER, MONO/
File: 05_master_fader.kicad_sch

***Title: MASTER FADER, MONO***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 5/30 |

MAIN OUTPUTS: DRV134 -> 10u NP in the Sense loop -> mute relay (NO; NC grounds the line) -> XLR male, pin 2 hot



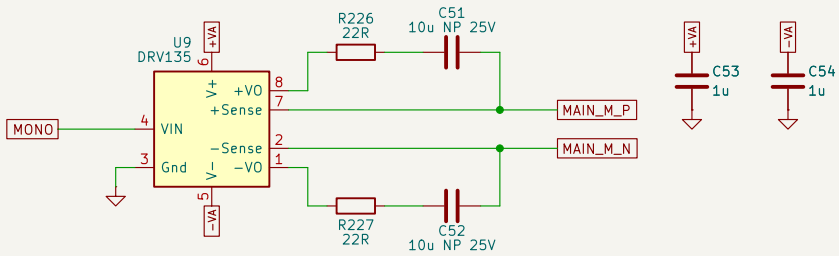

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /06 MAIN OUTPUTS/
File: 06_main_outputs.kicad_sch

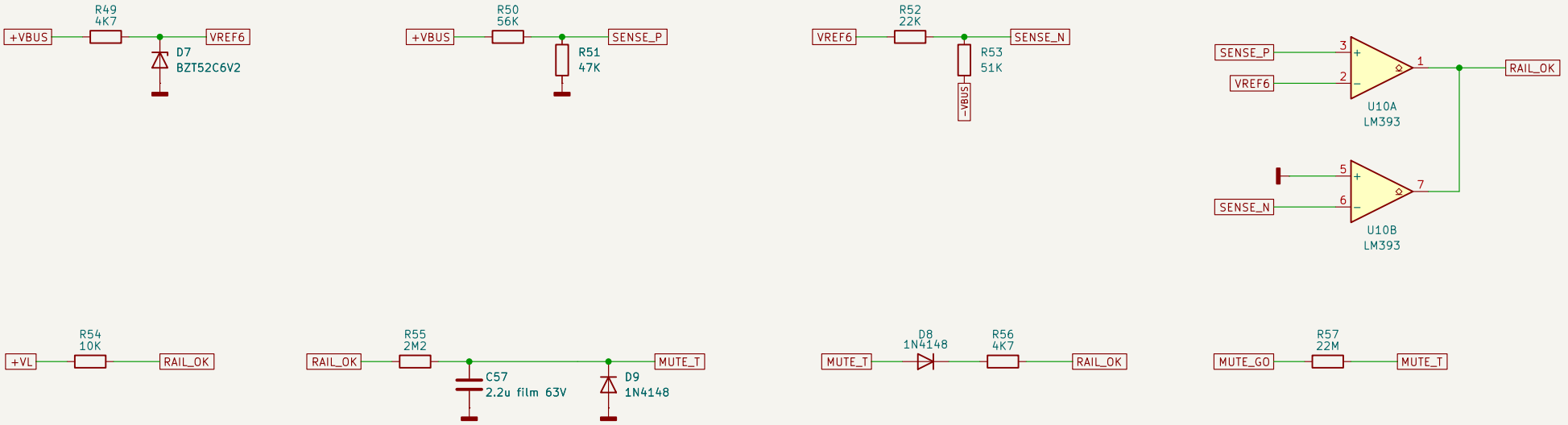
**Title: MAIN OUTPUTS**

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 6/30 |

## MAIN OUTPUTS: DRV134 -> 10u NP in the Sense loop -> mute relay (NO; NC grounds the line) -> XLR male, pin 2 hot (cont.)




## POWER-ON MUTE: rails > ~14 V for ~2.7 s closes all six relays; either rail low opens them at once



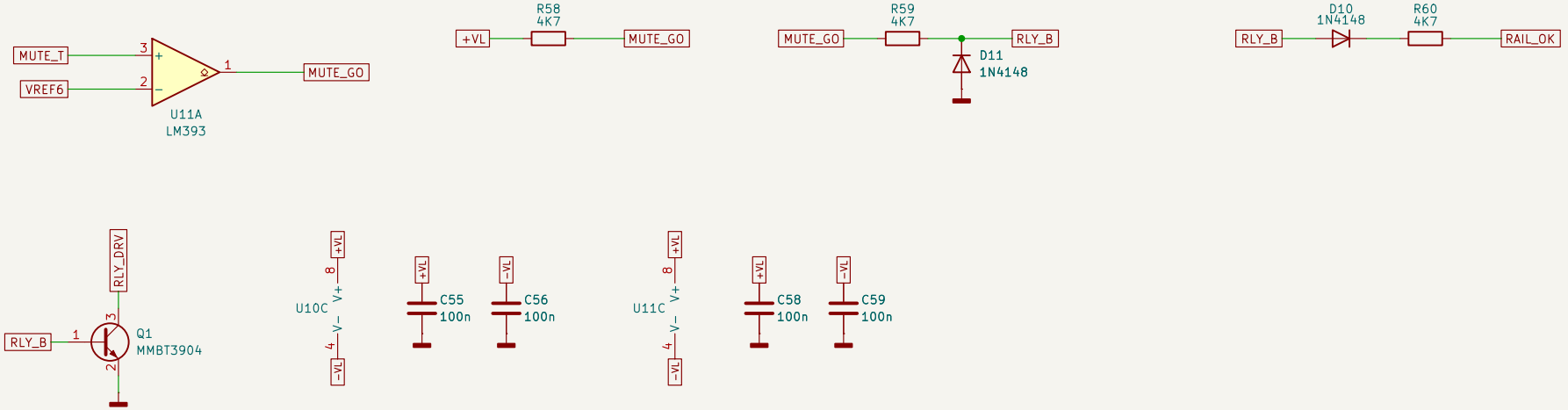

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /07 MAIN OUTPUTS, POWER-ON MUTE/
File: 07_main_outputs.kicad_sch

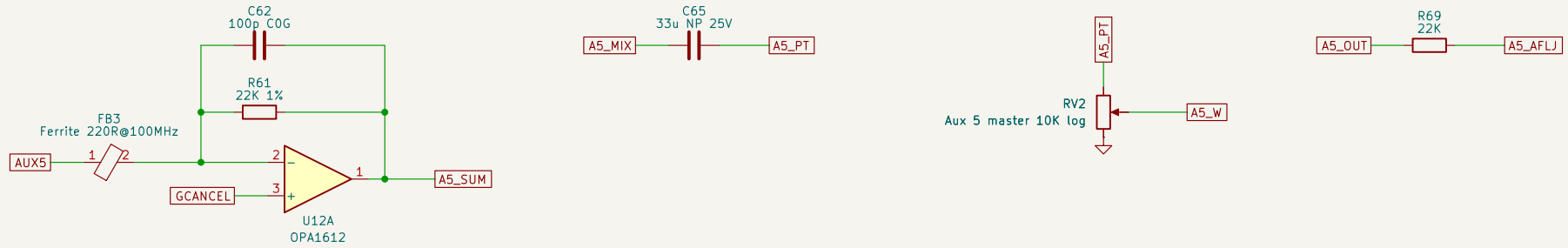
**Title: MAIN OUTPUTS, POWER-ON MUTE**

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 7/30 |

## POWER-ON MUTE: rails > ~14 V for ~2.7 s closes all six relays; either rail low opens them at once (cont.)




## AUX MASTER 5: OPA1612 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL



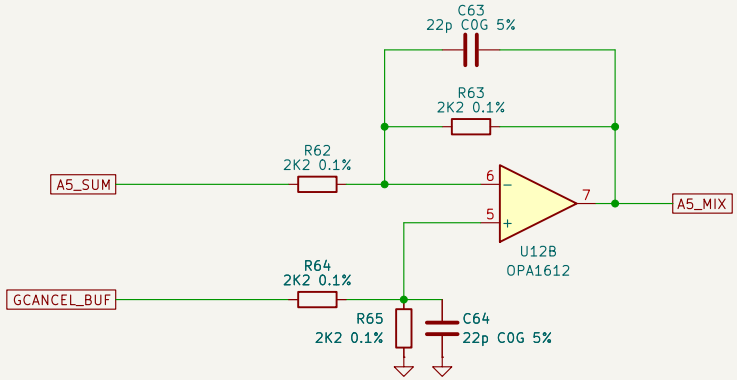

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

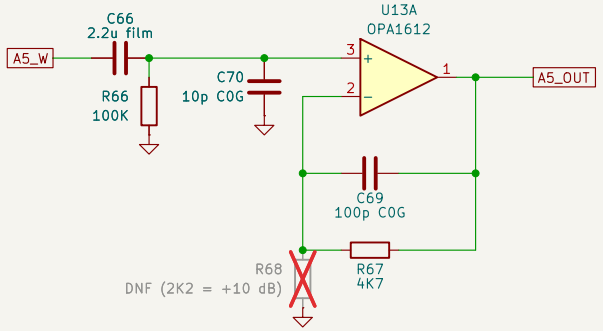
Sheet: /08 POWER-ON MUTE, AUX MASTER 5/
File: 08_power_on_mute.kicad_sch

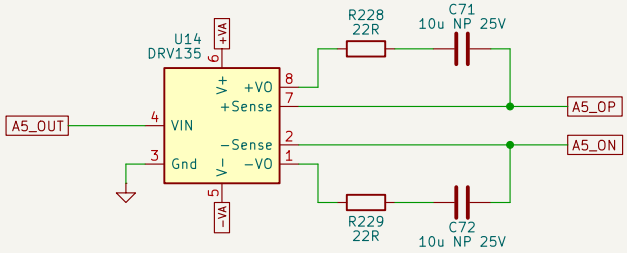
***Title: POWER-ON MUTE, AUX MASTER 5***

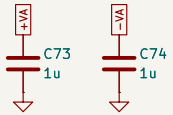
| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 8/30 |

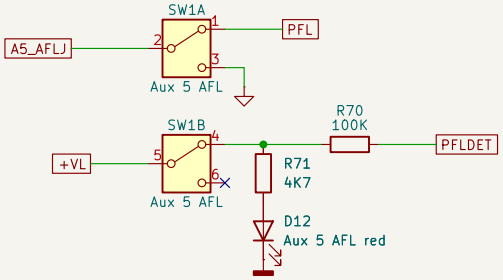
AUX MASTER 5: OPA1612 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL (cont.)

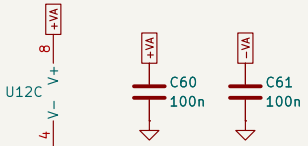


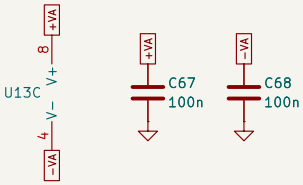



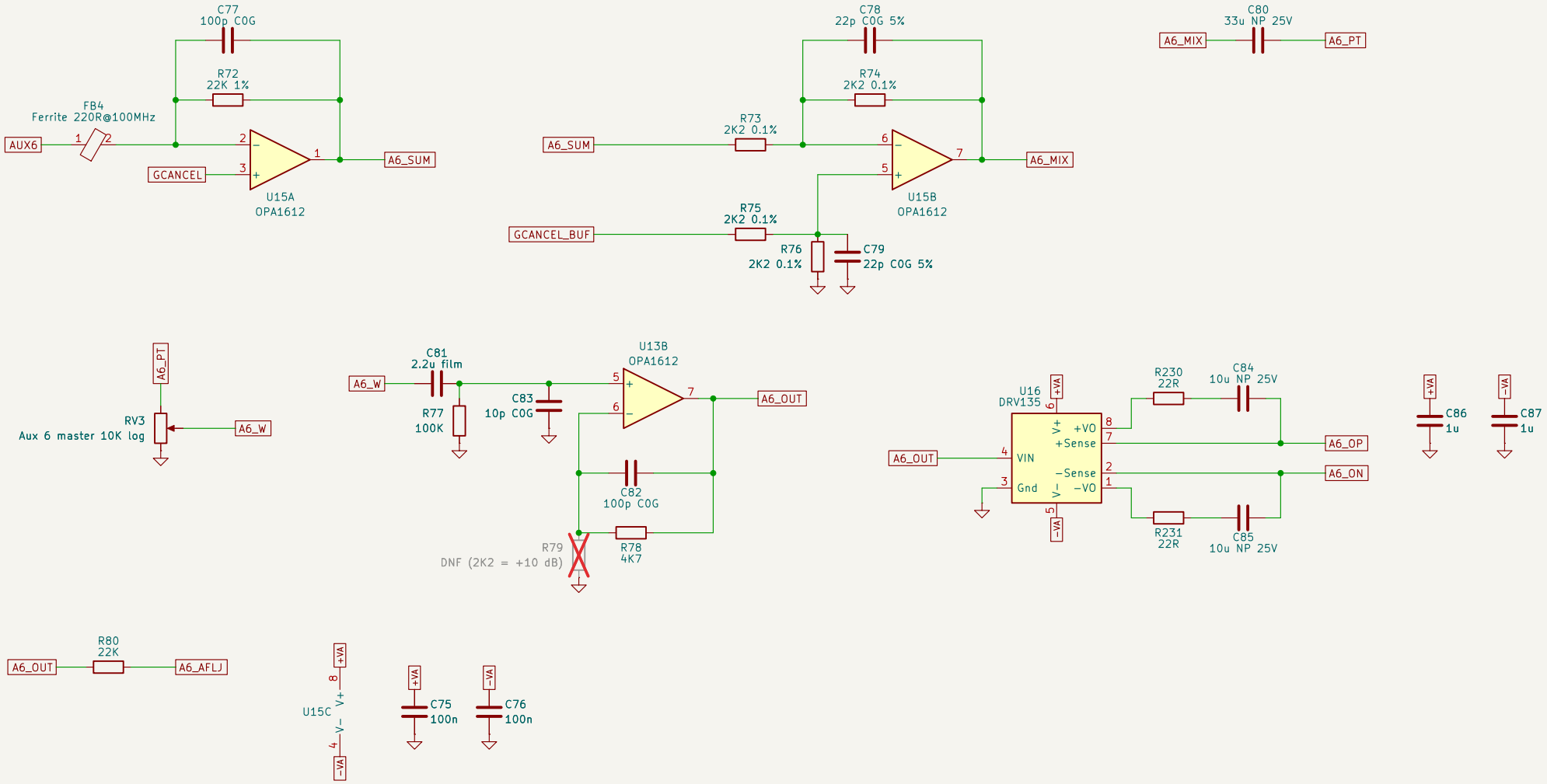

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /09 AUX MASTER 5/
File: 09_aux_master_5.kicad_sch

**Title: AUX MASTER 5**

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 9/30 |

AUX MASTER 6: OPA1612 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL



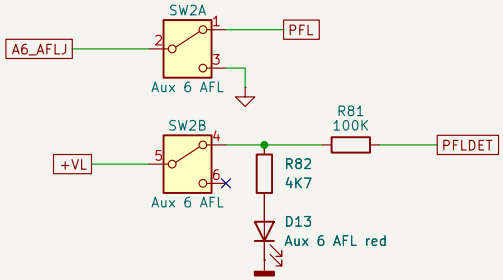

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /10 AUX MASTER 6/
File: 10_aux_master_6.kicad_sch

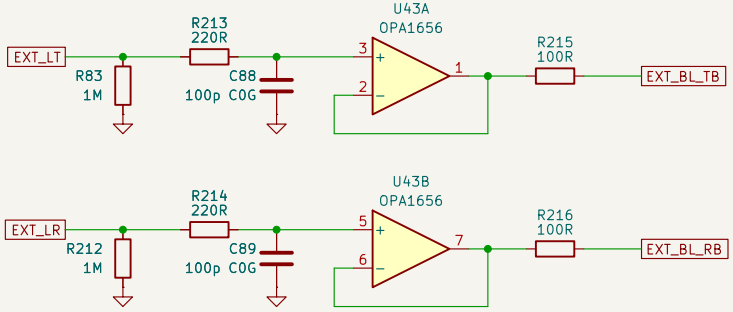
***Title: AUX MASTER 6***

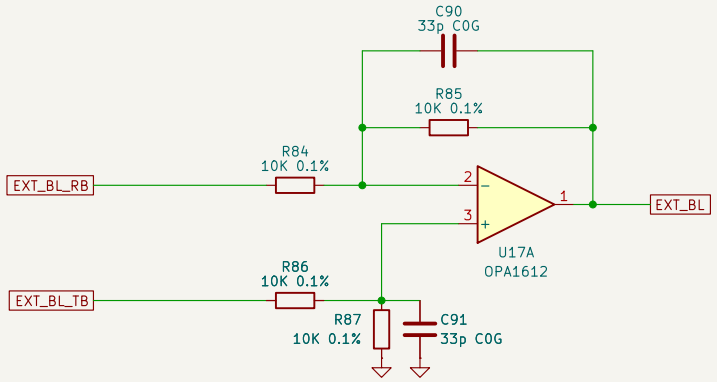
| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 10/30 |

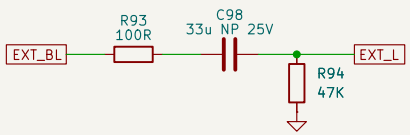
## AUX MASTER 6: OPA1612 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL (cont.)

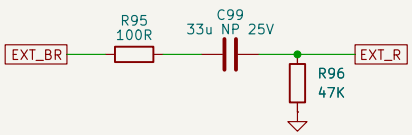



## EXT INPUT (2-track / DAW) and MIX / EXT monitor select  (SW latched = EXT)



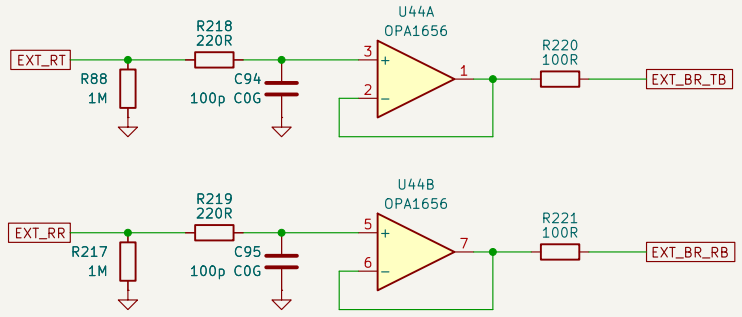

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

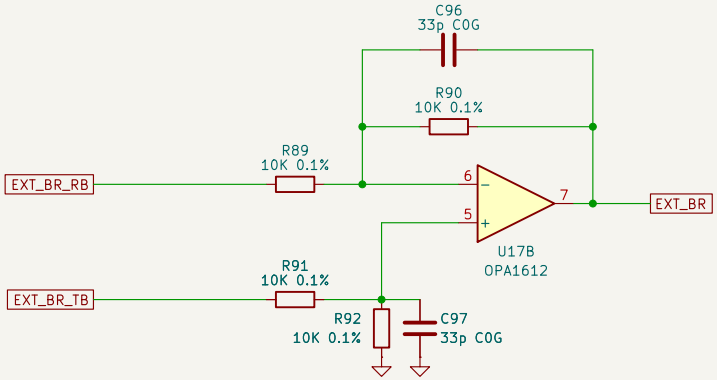
Sheet: /11 AUX MASTER 6, EXT INPUT/
File: 11_aux_master_6.kicad_sch

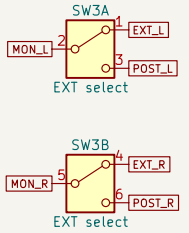
***Title: AUX MASTER 6, EXT INPUT***

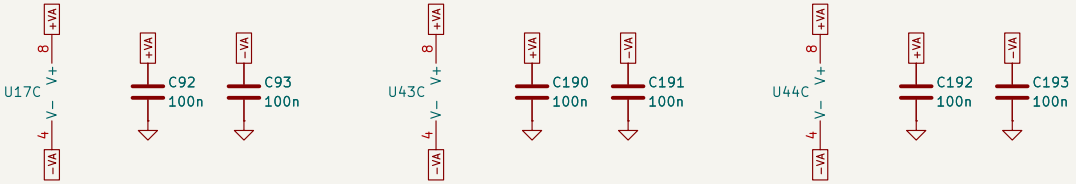
| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 11/30 |

## EXT INPUT (2-track / DAW) and MIX / EXT monitor select (cont.)

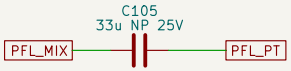


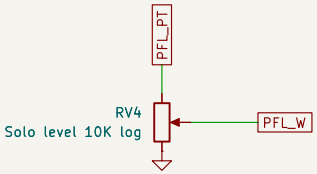



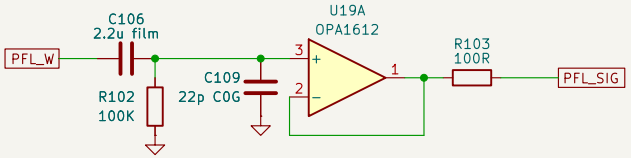



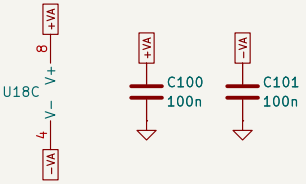




## PFL / AFL: OPA1656 sum (22K || 100p) + re-invert, SOLO LEVEL, follower




Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9

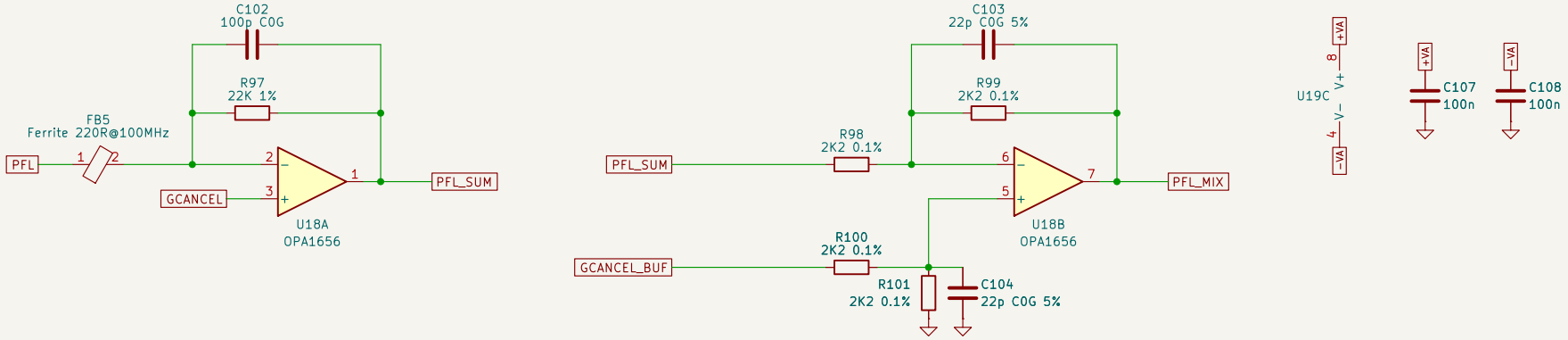
**Master section — goa console**

Sheet: /12 EXT INPUT, PFL-AFL/
File: 12_ext_input.kicad_sch

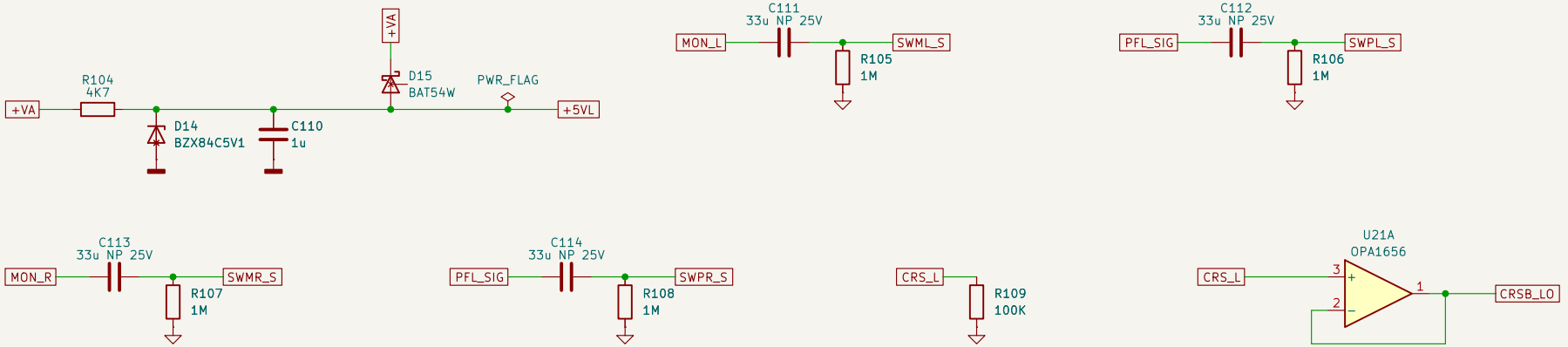
**Title: EXT INPUT, PFL-AFL**

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 12/30 |

PFL / AFL: OPA1656 sum (22K || 100p) + re-invert, SOLO LEVEL, follower (cont.)




PFL SWITCHER: DG413 (PFL pressed = SW1/SW4, else SW2/SW3; break-before-make) -> CRS L/R OPA1656 followers



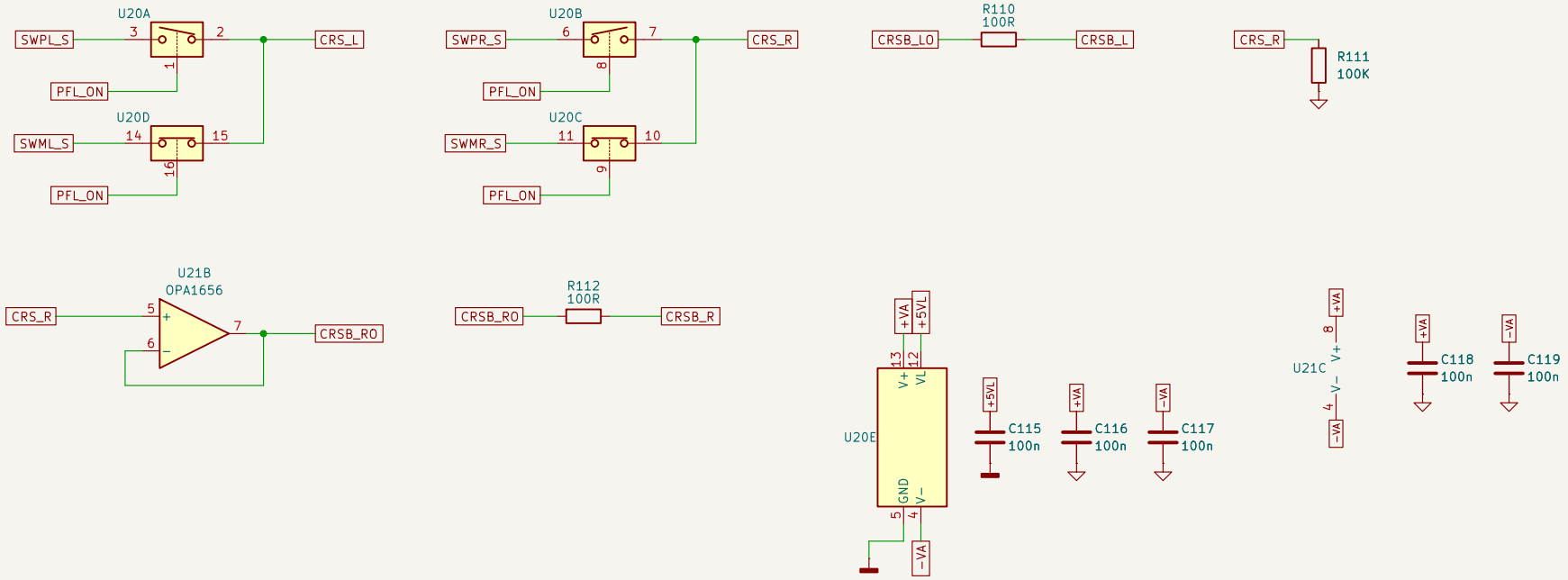

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /13 PFL-AFL, PFL SWITCHER/
File: 13_pfl_afl.kicad_sch

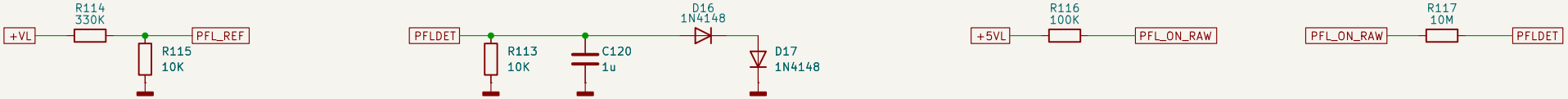
***Title: PFL-AFL, PFL SWITCHER***

| | | |
|---|---|---|
| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
| KiCad E.D.A. 10.0.7 | | Id: 13/30 |

PFL SWITCHER: DG413 (PFL pressed = SW1/SW4, else SW2/SW3; break-before-make) -> CRS L/R OPA1656 followers (cont.)




PFL DETECT (0.92 V per press vs 0.50 V; 2 x 1N4148 clamp, 1u debounce) and DG413 logic, beside the switcher




Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /14 PFL SWITCHER, PFL DETECT/
File: 14_pfl_switcher.kicad_sch

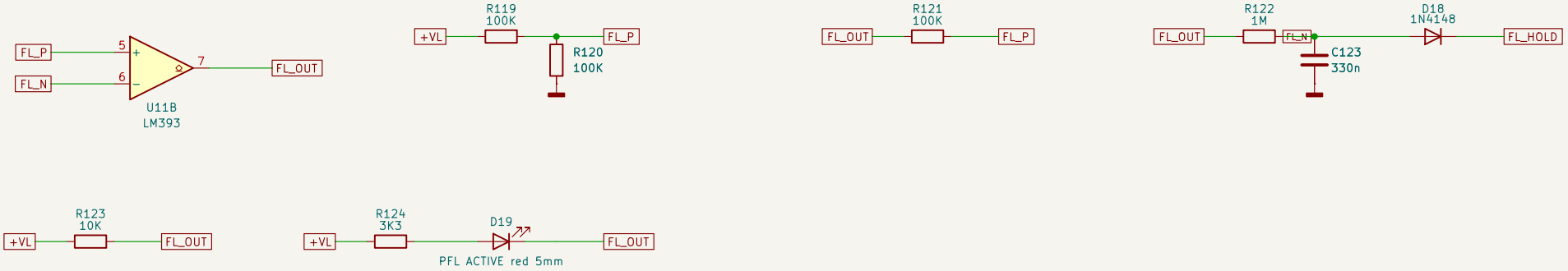
***Title: PFL SWITCHER, PFL DETECT***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 14/30 |

PFL DETECT (0.92 V per press vs 0.50 V; 2 x 1N4148 clamp, 1u debounce) and DG413 logic, beside the switcher (cont.)




RUDE SOLO: ~2 Hz flasher, PFL ACTIVE LED; held off through FL_N while no key is pressed



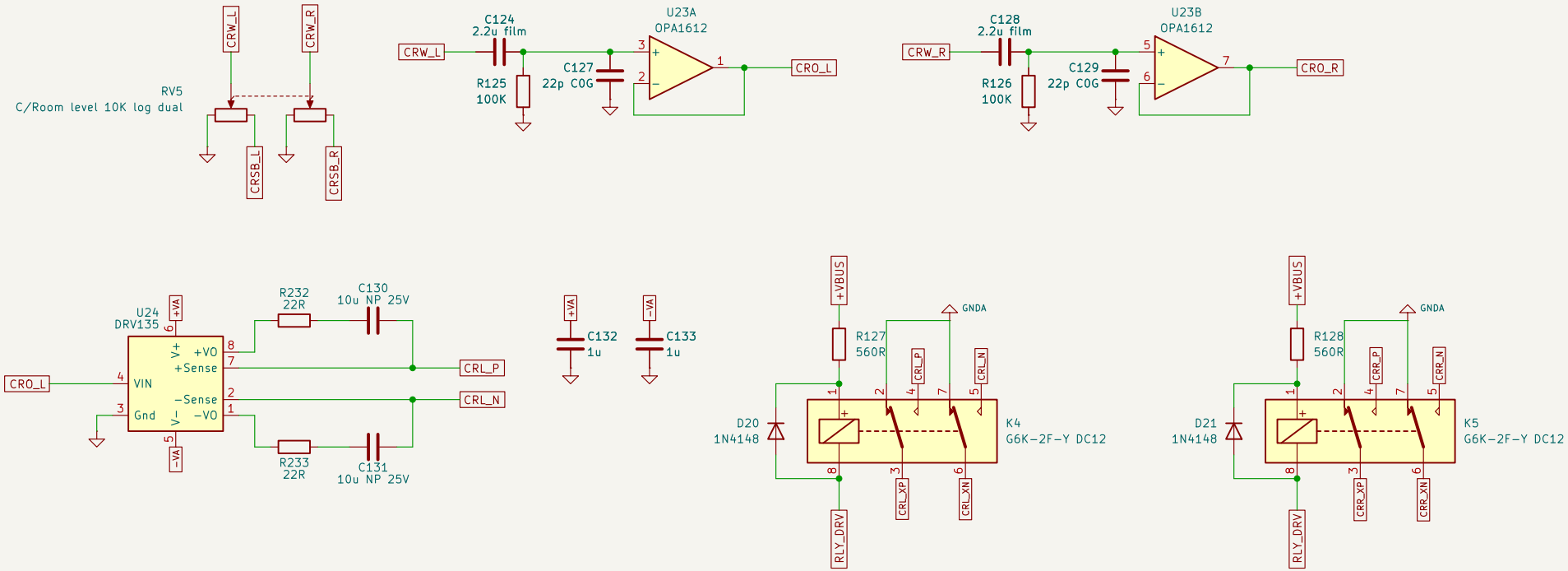

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /15 PFL DETECT, RUDE SOLO/
File: 15_pfl_detect.kicad_sch

***Title: PFL DETECT, RUDE SOLO***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 15/30 |

CONTROL ROOM: C/ROOM dual A10K, followers, DRV134, mute relays, TRS outs



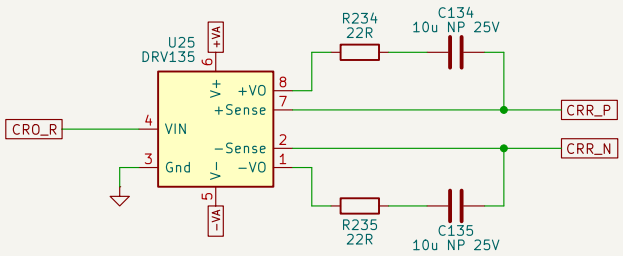

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

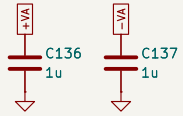
Sheet: /16 CONTROL ROOM/
File: 16_control_room.kicad_sch

***Title: CONTROL ROOM***

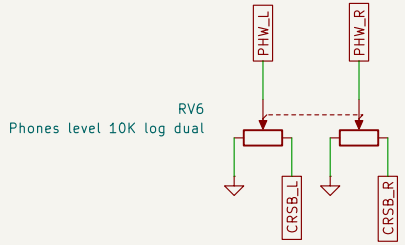
| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 16/30 |

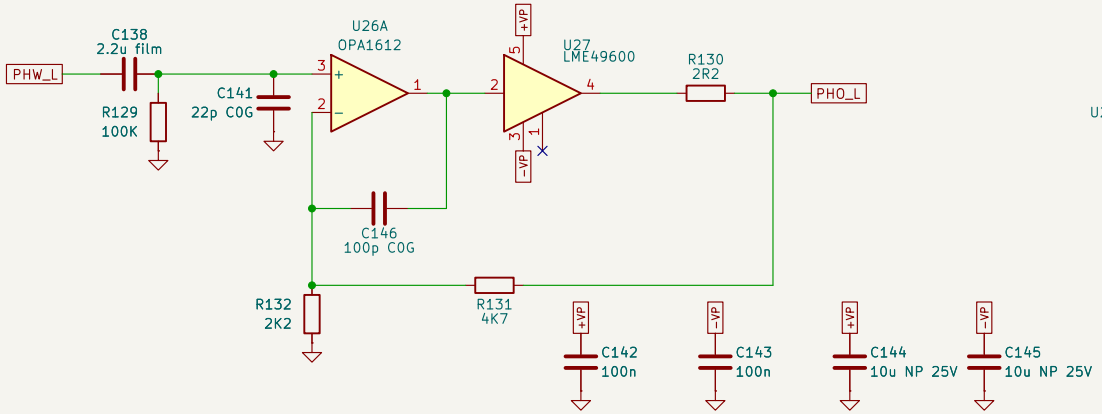
## CONTROL ROOM: C/ROOM dual A10K, followers, DRV134, mute relays, TRS outs (cont.)

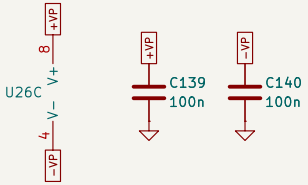



## HEADPHONES: PHONES dual A10K, OPA1612 + LME49600 composite (+9.9 dB), 2R2 out, mute relay, PHONESGND







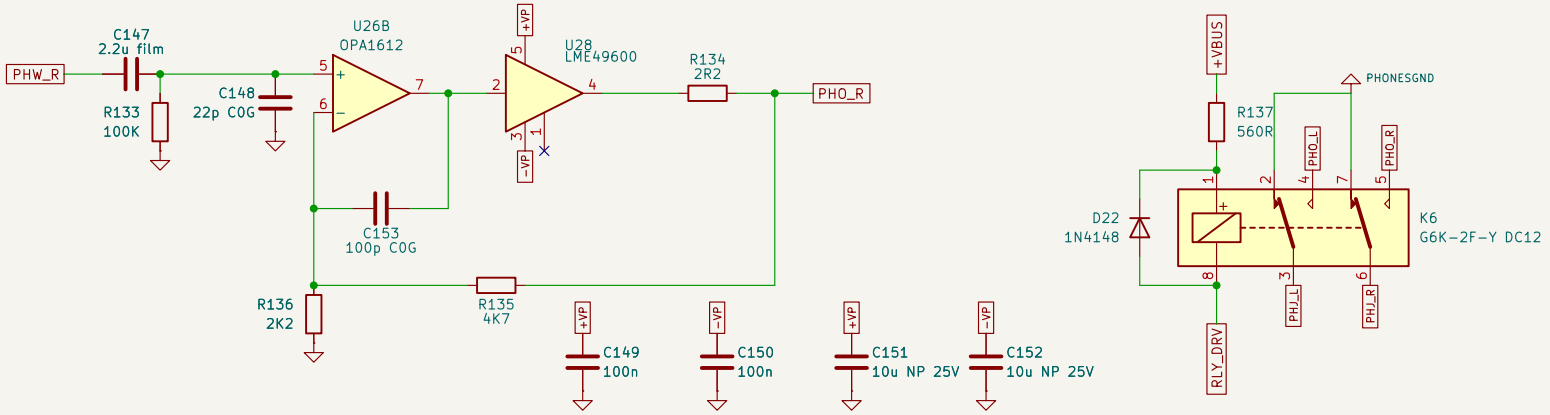
Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /17 CONTROL ROOM, HEADPHONES/
File: 17_control_room.kicad_sch

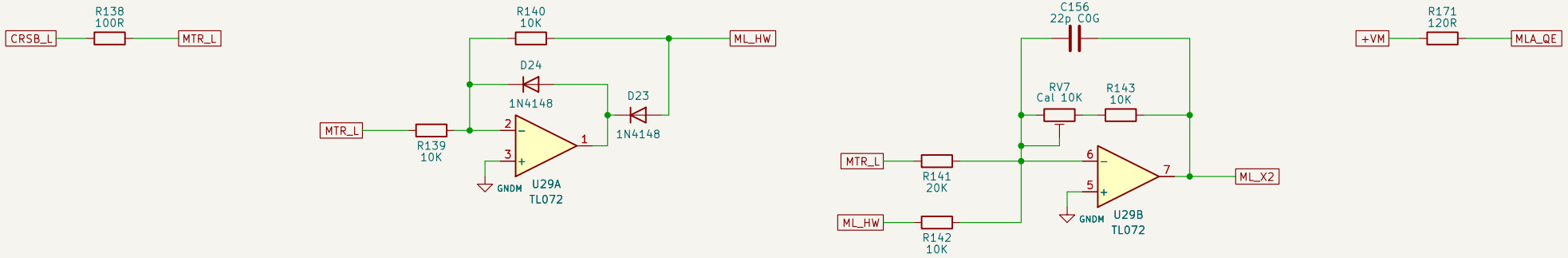
***Title: CONTROL ROOM, HEADPHONES***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 17/30 |

## HEADPHONES: PHONES dual A10K, OPA1612 + LME49600 composite (+9.9 dB), 2R2 out, mute relay, PHONESGND (cont.)




## METER L: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings




Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9

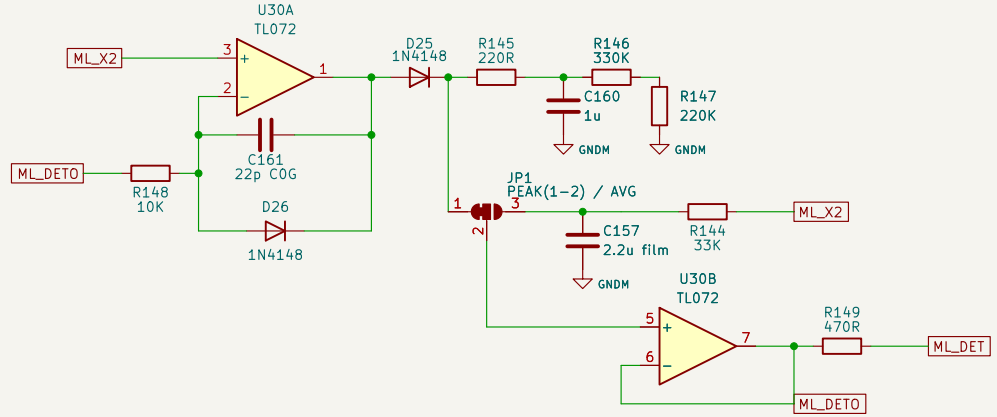
**Master section — goa console**

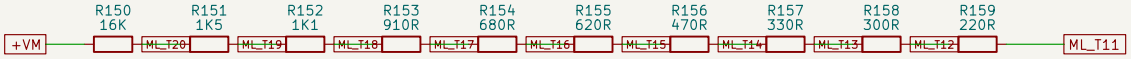
Sheet: /18 HEADPHONES, METER L/
File: 18_headphones.kicad_sch

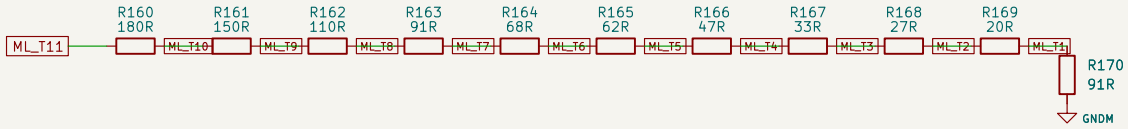
**Title: HEADPHONES, METER L**

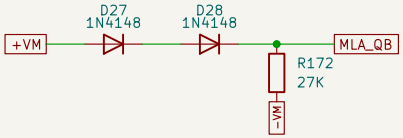
| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 18/30 |

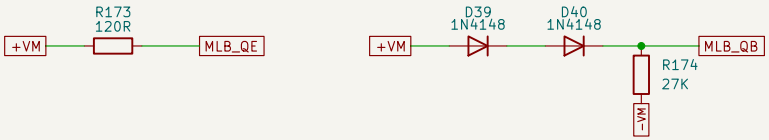
METER L: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)



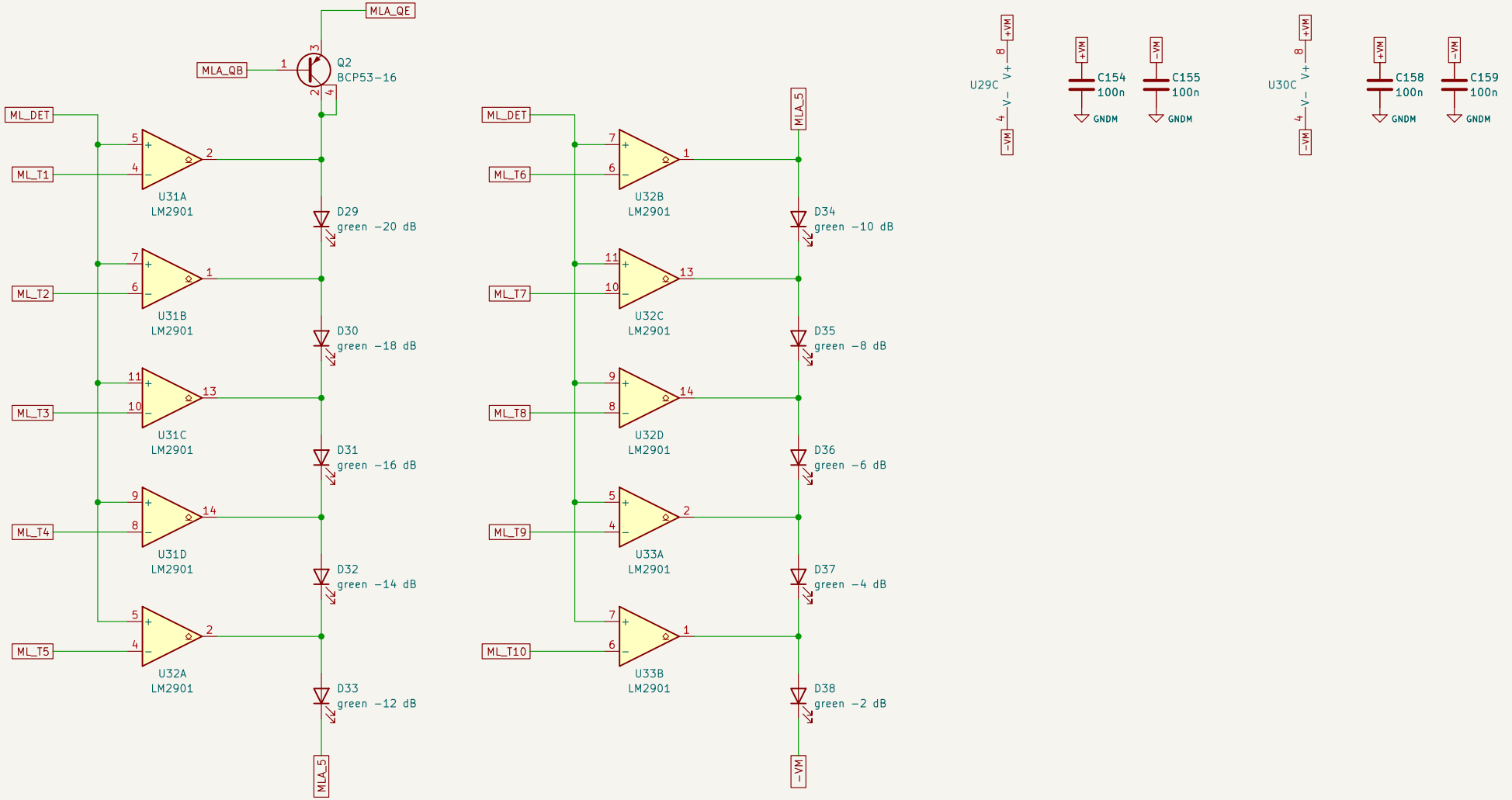

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /19 METER L/
File: 19_meter_l.kicad_sch

***Title: METER L***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 19/30 |

METER L: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)



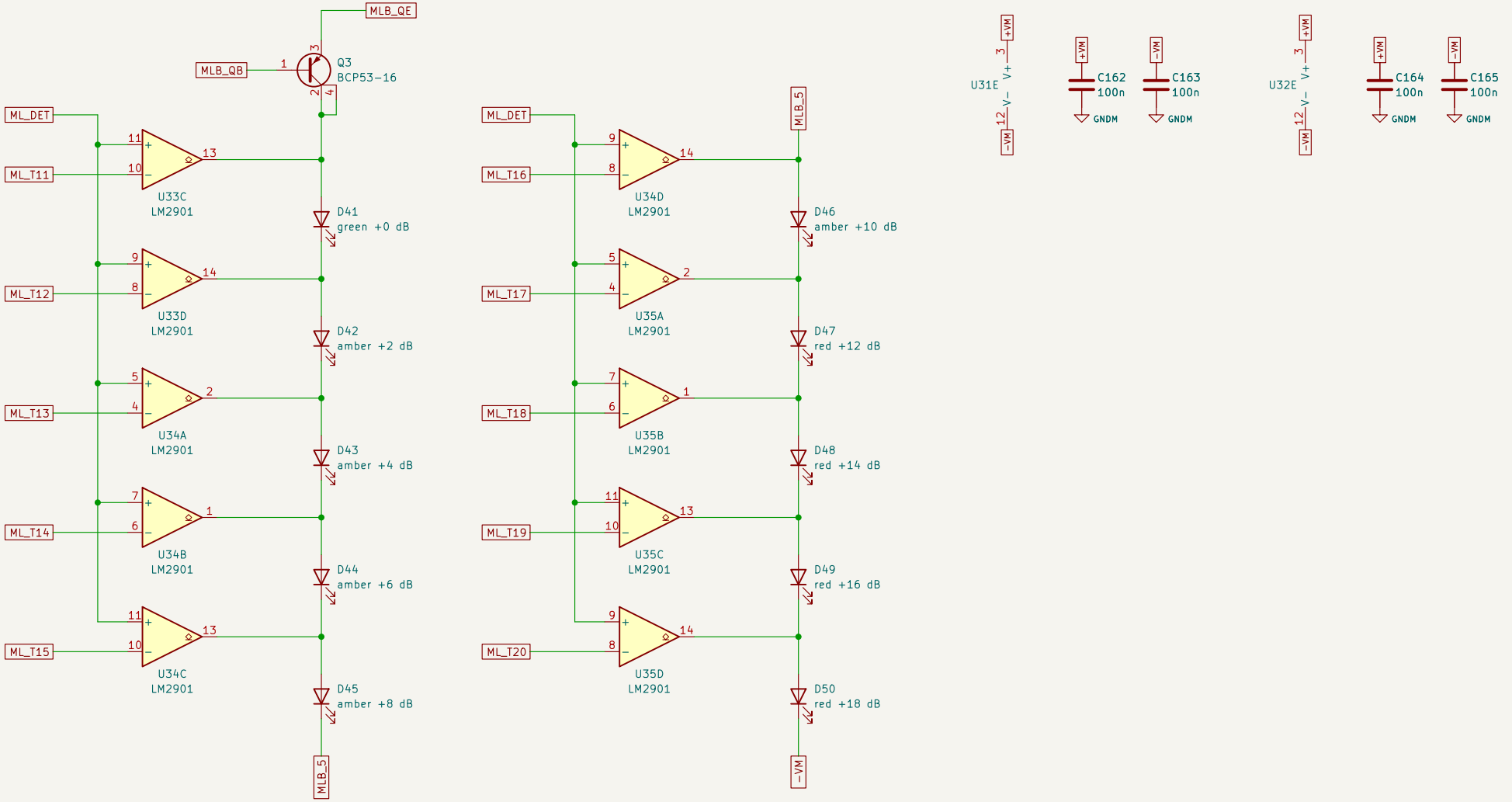

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /20 METER L/
File: 20_meter_l.kicad_sch

***Title: METER L***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 20/30 |

METER L: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)



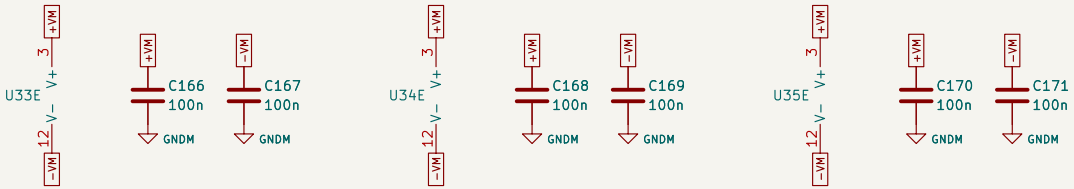

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /21 METER L/
File: 21_meter_l.kicad_sch

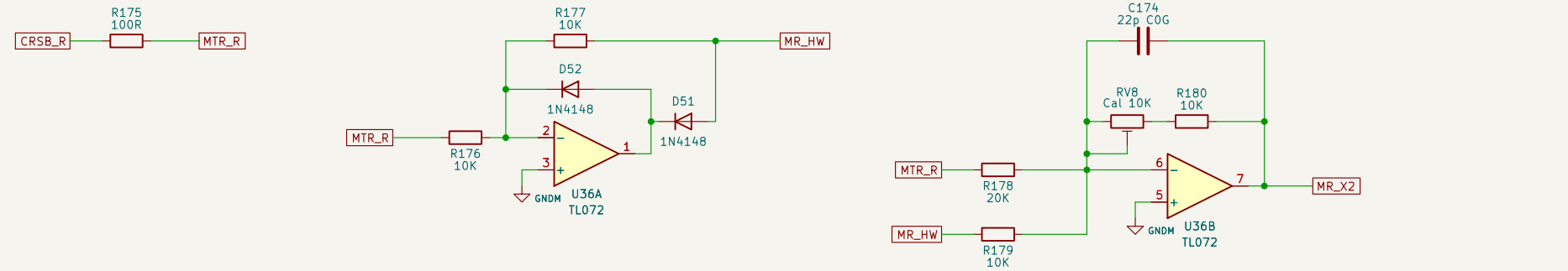
***Title: METER L***

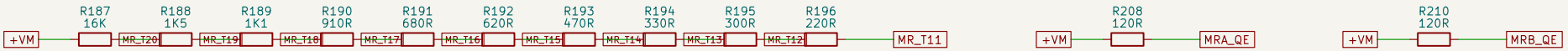
| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 21/30 |

## METER L: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)




## METER R: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings



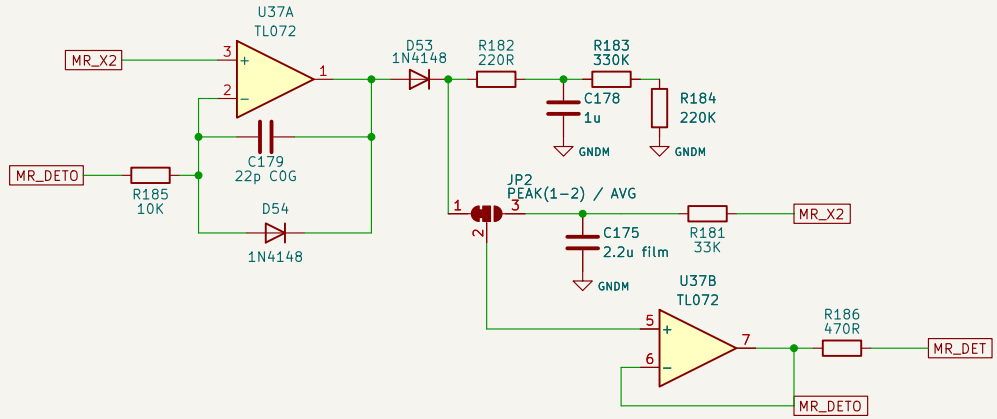

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

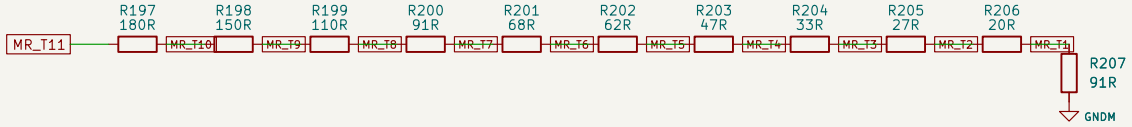
Sheet: /22 METER L, METER R/
File: 22_meter_l.kicad_sch

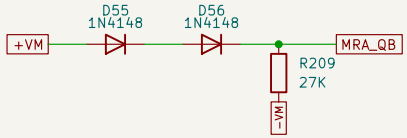
***Title: METER L, METER R***

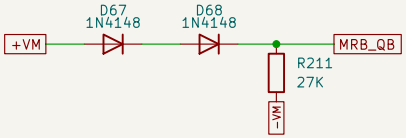
| | | |
|---|---|---|
| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
| KiCad E.D.A. 10.0.7 | | Id: 22/30 |

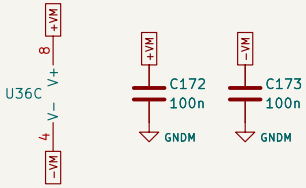
METER R: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)

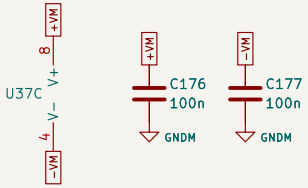


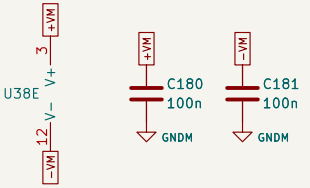



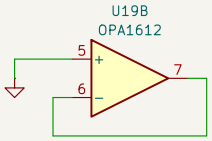



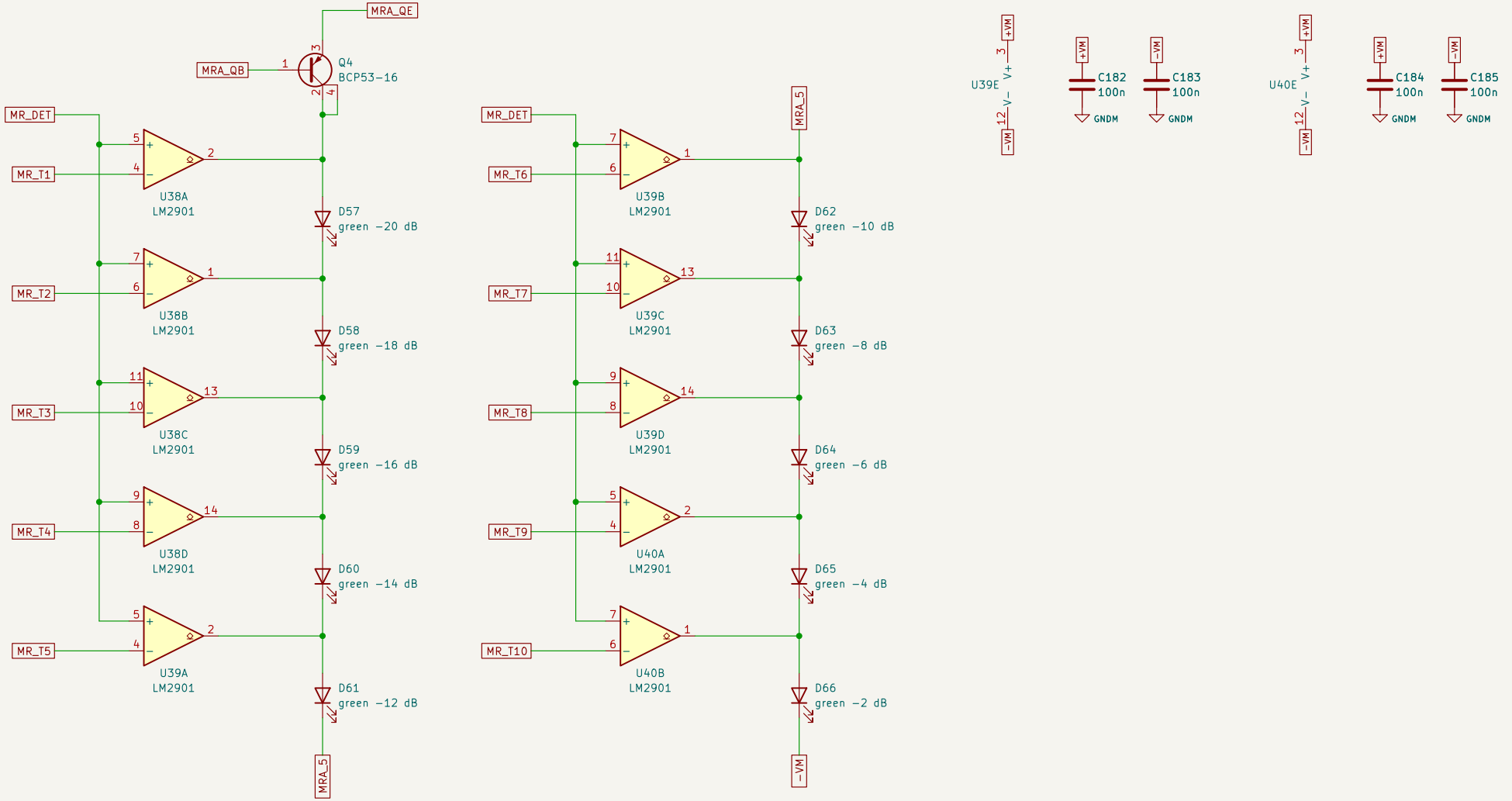

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /23 METER R/
File: 23_meter_r.kicad_sch

***Title: METER R***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 23/30 |

## METER R: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)



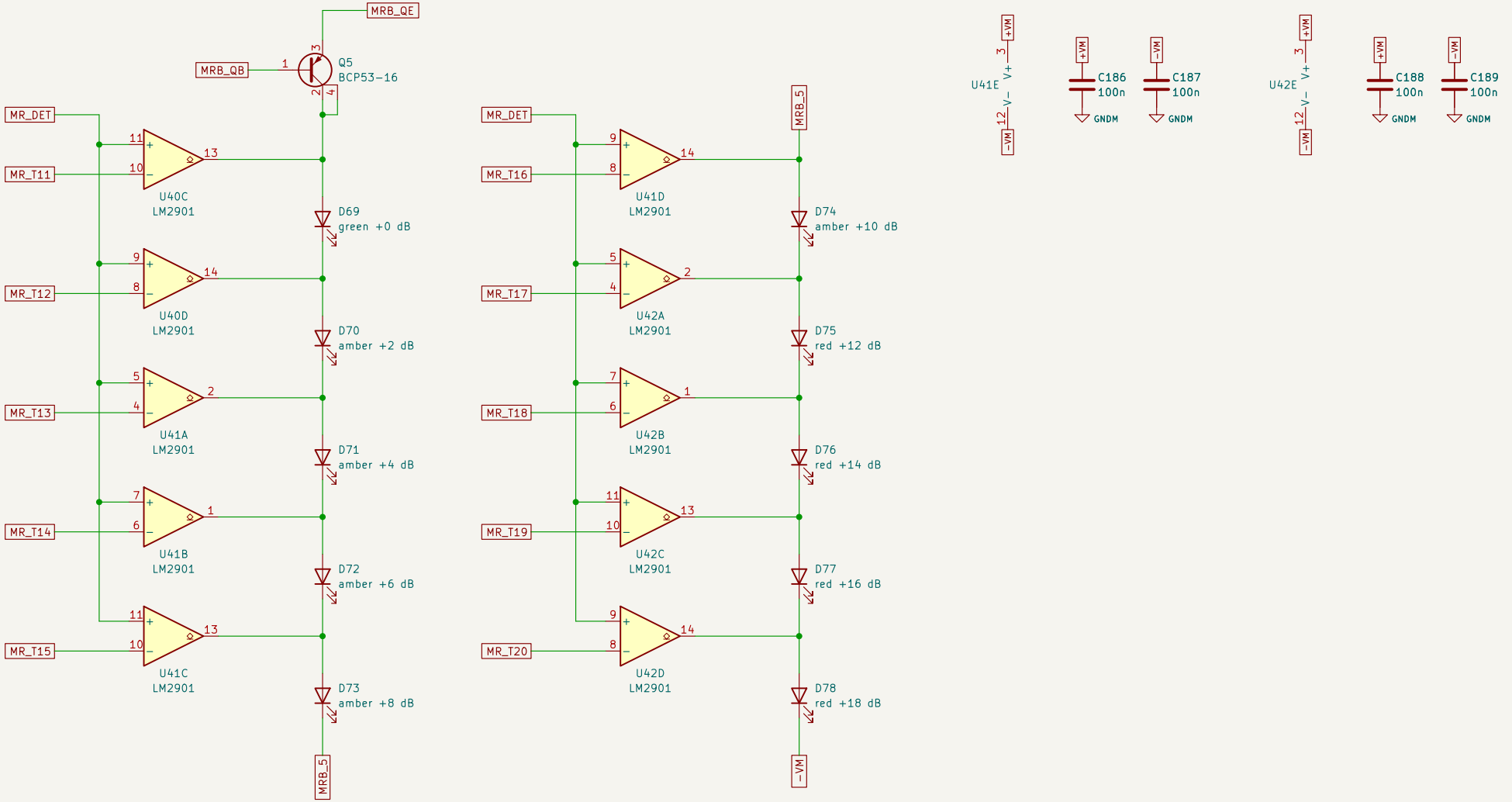

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /24 METER R/
File: 24_meter_r.kicad_sch

***Title: METER R***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 24/30 |

METER R: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings (cont.)



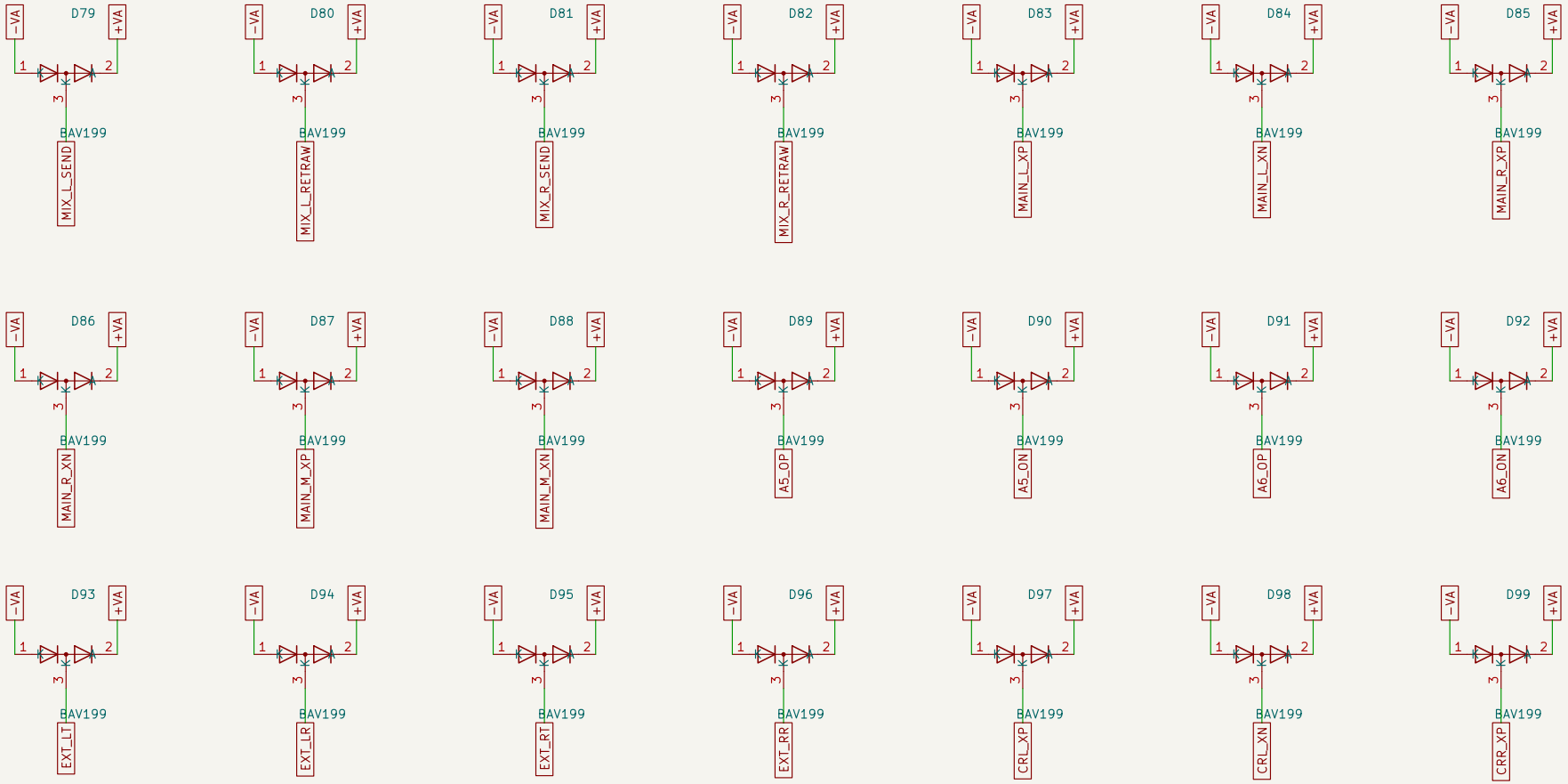

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /25 METER R/
File: 25_meter_r.kicad_sch

***Title: METER R***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 25/30 |

JACK LINK CLAMPS: a BAV199 to the audio rails on every jack signal at the link (CS4b)



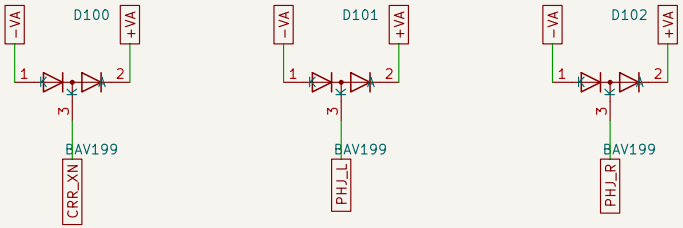

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /26 JACK LINK CLAMPS/
File: 26_jack_link_clamps.kicad_sch

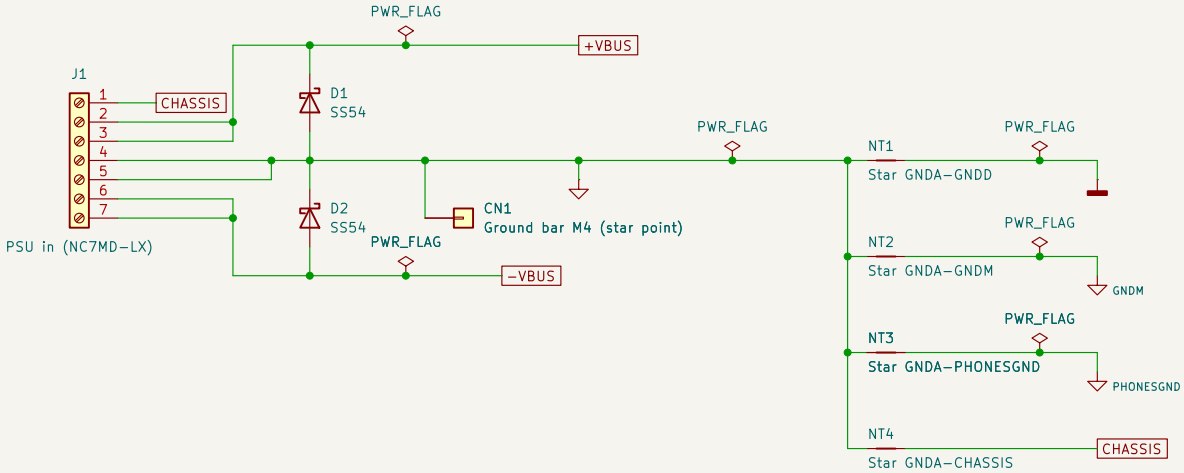
**Title: JACK LINK CLAMPS**

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 26/30 |

## JACK LINK CLAMPS: a BAV199 to the audio rails on every jack signal at the link (CS4b) (cont.)




## POWER ENTRY  (7-pin XLR NC7MD-LX on the rear panel, two 5 A contacts per rail, 18 AWG to this terminal; +-17 V PSU)



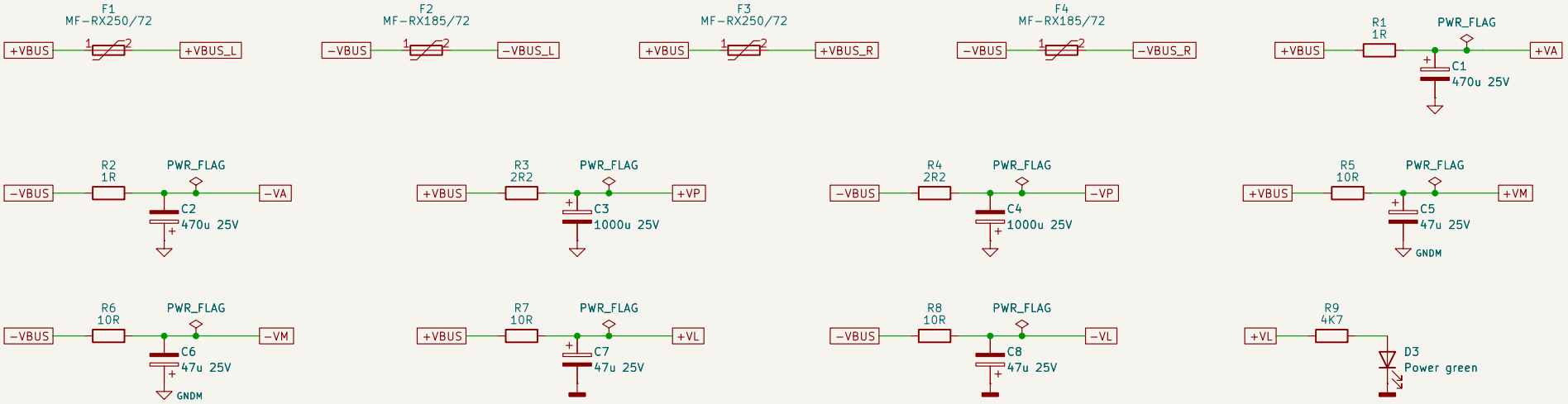

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /27 JACK LINK CLAMPS, POWER ENTRY/
File: 27_jack_link_clamps.kicad_sch

**Title: JACK LINK CLAMPS, POWER ENTRY**

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 27/30 |

DISTRIBUTION  (PTC per wing per rail, 1.85 A hold, 72 V; local branches VA audio, VP phones, VM meters, VL logic)



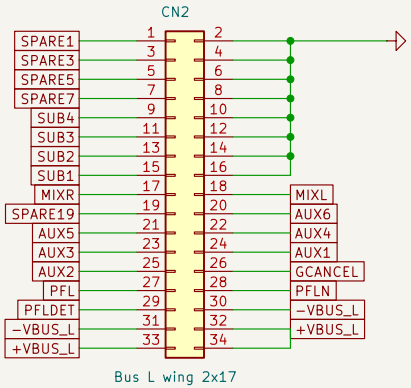

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

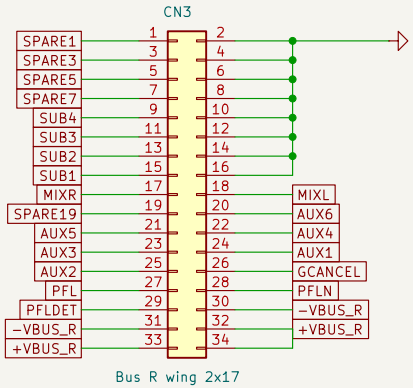
Sheet: /28 DISTRIBUTION/
File: 28_distribution.kicad_sch

***Title: DISTRIBUTION***

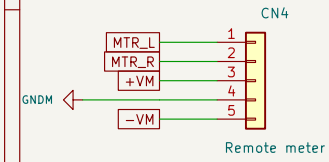
| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 28/30 |

## MOTHERLOOM (master in the centre: CN1 = left wing, CN2 = right wing; Delta pinout)




## REMOTE METER HEADER (for a meter bridge)



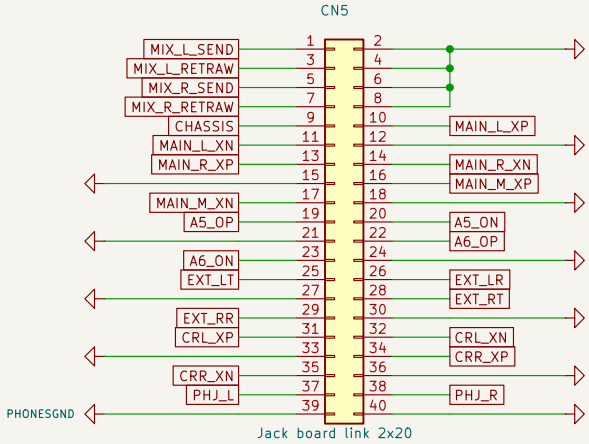

Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /29 MOTHERLOOM, REMOTE METER HEADER/
File: 29_motherloom.kicad_sch

***Title: MOTHERLOOM, REMOTE METER HEADER***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 29/30 |

JACK BOARD LINK  2x20 IDC ribbon to the master jack board (9 jacks, 3 XLRs)




Drawn by the master_lib fragments (tools/frags.py); refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9
**Master section — goa console**

Sheet: /30 JACK BOARD LINK/
File: 30_jack_board_link.kicad_sch

***Title: JACK BOARD LINK***

| Size: A4 | Date: 2026-10-19 | **Rev: 0.9** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 30/30 |